

Prajapati Dhruv

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EDUCATION

LDRP Institute of Technology and Research

Bachelor of Engineering — Information Technology

Gandhinagar, Gujarat

Aug. 2022 – May 2027

PROJECTS

Crowd Surveillance Anomaly Detection | *Python, YOLOv8, MediaPipe, OpenCV*

2024 – 2025

- Built a real-time crowd violence detection pipeline combining YOLOv8 Nano person detection, DeepSORT multi-object tracking, and MediaPipe pose estimation; achieved **88% accuracy** and **80% recall** on 50 labelled real-world clips.
- Designed a 3-rule alert fusion system — velocity (rolling 5-frame avg >15 px/frame), bounding-box IoU overlap (>0.20), and arm-elevation pose flag — requiring all conditions to hold for 3 consecutive frames, reducing false positive rate to **4%**.
- Implemented CUDA-accelerated YOLO inference achieving **15–20 FPS** on CPU; added a 2-second rolling pre-incident buffer that auto-saves alert clips, structured per-alert logging, and a modular 7-file architecture.

IPL Powerplay Aggression Predictor | *Python, XGBoost, SHAP, Flask, Pandas, Scikit-learn*

2025 – 2026

- Built an end-to-end ML system to predict IPL powerplay batting aggression using **17 hand-engineered features** derived from **1,169 IPL matches** (2008–2025) enriched with **10,984 T20 matches** across 20 global tournaments (BBL, PSL, CPL, T20I).
- Trained XGBoost regressors with walk-forward cross-validation across 5 held-out seasons (2021–2025), achieving mean aggression MAE of **4.87 ± 0.55 percentage points** — an honest out-of-sample evaluation methodology.
- Integrated SHAP explainability to surface per-feature contributions for each prediction; delivered a Flask web app with a prediction interface, team leaderboard, and venue analytics dashboard.

TECHNICAL SKILLS

Languages: Python, C++

ML / CV Libraries: YOLOv8 (Ultralytics), OpenCV, MediaPipe, PyTorch, XGBoost, Scikit-learn, SHAP, Pandas, NumPy

Web / Backend: Flask

Tools & Platforms: Git, VS Code, CUDA, Jupyter Notebook

CERTIFICATIONS & COURSES

Python for Data Science

Online Certification Course

NPTEL (IIT Madras)

Python (Basic) Certificate

Skill Certification

HackerRank